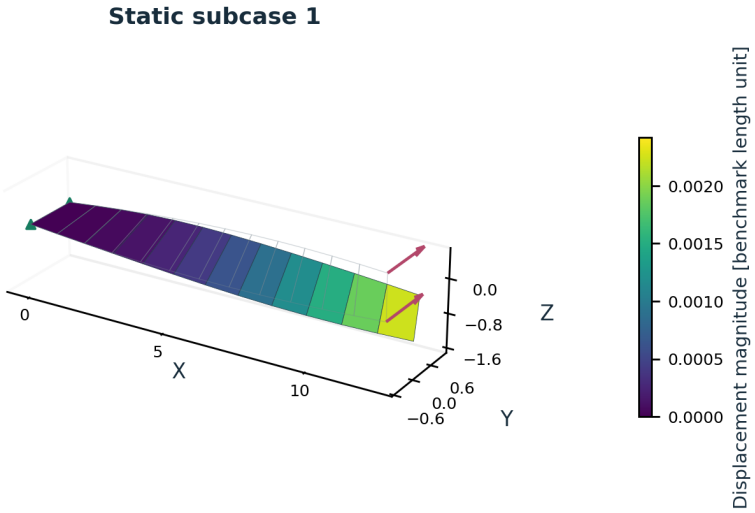
Twisted cantilever: transverse tip load

MH_twisted_beam_out_of_plane | SOL 101 | Published shell benchmark

Same 90-degree twisted rectangular shell as in-plane companion.

Geometry / mesh	26 nodes; 12 elements. Same 90-degree twisted rectangular shell as in-plane companion.
Material / section	E=2.9e+07, nu=0.22; shell thickness=0.32. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Root clamped; unit total out-of-plane tip force along global Y.
Applied load	Total Fy=1 at the twisted free section (local out-of-plane bending).



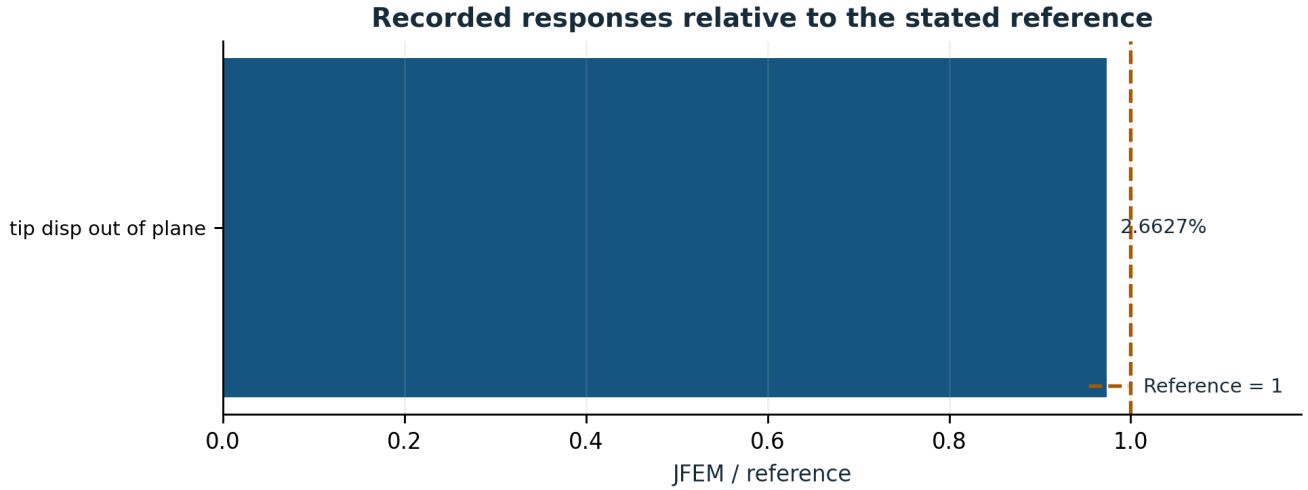
Shape amplified x595; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
tip disp out of plane [benchmark length unit]	0.0017073	0.001754	2.6627%	PASS

Evidence, convergence and interpretation



Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Ko, Y.; Lee, Y.; Lee, P.-S.; Bathe, K.-J. (2017). Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Computers & Structures 193, 187-206. https://doi.org/10.1016/j.compstruc.2017.08.003

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv